

Microelectronics

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1 Microelectronics

I Signal

i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems on voltage or current signals, especially the former. Only voltage and current signals are studied here.

ii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

iii Transducer

A transducer is a device that convert a signal to a voltage or current signal or vice versa.

iv Signal processing

Signal processing includes analog signal processing (ASP), analog-to-digital converter (ADC), digital signal processor (DSP), digital-to-analog converter (DAC), amplifier, etc.

v Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g. $i_C(t)$.
- Lowercase character with lowercase subscript means AC signal, e.g. $i_c(t)$.
- Uppercase character with uppercase subscript means DC signal, e.g. I_C .
- Uppercase character with lowercase subscript means peak signal, e.g. I_c .

Their relation is

$$i_C(t) = i_c(t) + I_C$$

$$i_c(t) = I_c \sin(\omega t + \varphi)$$

II Amplifier

i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain A and optionally convert voltage signal to current signal or vice versa.

ii Circuit model of voltage amplifier

A voltage amplifier draws current from a source v_s with internal resistance R_s with input resistance R_i and voltage v_{in} across it and supplies $v_c = \mu v_{in}$ to a load network with output resistance R_o and load R_L .

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$
$$\frac{v_c}{v_{in}} = \mu$$
$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu.$$

iii Circuit model of current amplifier

A current amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $i_c = \beta i_{in}$ by CCCS to a load network with output resistance R_o in parallel and load R_L .

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$
$$\frac{i_c}{i_{in}} = \beta$$
$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta.$$

iv Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source v_s with internal resistance R_s in series with input resistance R_i and current i_{in} through it and supplies $i_c = g v_{in}$ by VCCS to a load network with output resistance R_o in parallel and load R_L .

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

2

$$\frac{i_c}{v_{in}} = g$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g \frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g.$$

v Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $v_c = r i_{in}$ by CCVS to a load network with output resistance R_o in series and load R_L .

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{v_c}{i_{in}} = r$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = \beta \frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r.$$

vi Amplifier gain in decibels (dB)

A power gain A_p in dB is defined as

$$G_p = 10 \log_{10}(A_p).$$

A voltage gain A_v in dB is defined as

$$G_v = 20 \log_{10}(A_v).$$

A current gain A_c in dB is defined as

$$G_c = 20 \log_{10}(A_c).$$

vii Amplifiers in cascade

Output of an amplifier as input of another amplifier.

III Operational Amplifier (op amp or op-amp)

i Definition

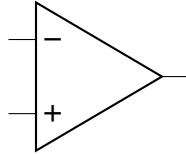
An op amp has five terminals:

- v_+ : non-inverting input voltage of the terminal at the + sign;
- v_- : inverting input voltage of the terminal at the – sign;
- v_o : output voltage of the terminal at the opposite side of inputs;
- v_{CC+} : positive supply voltage, denoted on the + side of an op-amp or omitted;
- v_{CC-} : negative supply voltage, denoted on the – side of an op-amp or omitted.

$$v_o = A(v_+ - v_- - v_{os}),$$

where A is open-loop gain, and v_{os} is input offset voltage.

ii Diagram



iii Ideal op amp

- Infinite open-loop gain, infinite bandwidth, and zero output impedance: $A = \infty$ and is independent of frequency and output voltage.
- Zero input offset voltage: $v_{os} = 0$.
- No current flow into either of the input terminals.

This is a linear element with inputs v_+ and v_- and output v_o .

iv General amplifier with inverting configuration

A general amplifier with inverting configuration consists of an op amp where input voltage v_i is connected to v_- through an input impedance Z_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback impedance Z_f .

$$I_i = \frac{V_i}{Z_i} = I_f = \frac{-V_o}{Z_f},$$

$$V_o = -\frac{Z_f}{Z_i} V_i.$$

When $\left| \Re\left(\frac{Z_f}{Z_i}\right) \right| < 1$, it's called an attenuator.

v Inverting amplifier

An inverting amplifier consists of an op amp where input voltage v_i is connected to v_- through an input resistor R_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f .

$$i_i = \frac{v_i}{R_i} = i_f = \frac{-v_o}{R_f},$$
$$v_o = -\frac{R_f}{R_i} v_i.$$

vi General amplifier with noninverting configuration

A general amplifier with noninverting configuration consist of an op amp where input voltage v_i is connected to v_+ , v_- is connected ground through a ground impedance Z_g , and v_o is fed back to v_- through a feedback impedance Z_f .

$$V_i = V_o \frac{Z_g}{Z_f + Z_g},$$
$$V_o = \left(1 + \frac{Z_f}{Z_g}\right) V_i.$$

vii Noninverting amplifier

A noninverting amplifier consist of an op amp where input voltage v_i is connected to v_+ , v_- is connected ground through a ground resistor R_g , and v_o is fed back to v_- through a feedback resistor R_f .

$$v_i = v_o \frac{R_g}{R_f + R_g},$$
$$v_o = \left(1 + \frac{R_f}{R_g}\right) v_i.$$

viii Unity gain amplifier, unity gain buffer, or voltage follower

A unity gain amplifier consist of an op amp where input voltage v_i is connected to v_+ and v_o is fed back to v_- .

$$v_o = v_i.$$

ix Summing amplifier

A summing amplifier consists of an op amp, input voltages $v_1, v_2, \dots, v_n$, n input resistors $R_1, R_2, \dots, R_n$, and a feedback resistor R_f , where for any $k \in \mathbb{N} \wedge k \leq n$, v_k is connected to v_- through R_k , v_+ is connected to ground, and v_o is fed back to v_- through R_f .

$$i_k = \frac{v_k}{R_k},$$
$$\sum_{k=1}^n i_k = i_f = \frac{-v_o}{R_f},$$

$$v_o = -R_f \sum_{k=1}^n \frac{v_k}{R_k}.$$

When $R_k = R$ for all $k \in \mathbb{N} \wedge k \leq n$:

$$v_o = -\frac{R_f}{R} \sum_{k=1}^n v_k.$$

x Difference amplifier or differential amplifier

A difference amplifier consists of a minuend v_1 , a subtrahend v_2 , and four resistors R_1, R_2, R_3, R_4 satisfying the condition

$$\frac{R_1}{R_2} = \frac{R_3}{R_4},$$

where v_1, v_2 are connected to v_-, v_+ through R_1, R_2 respectively, v_o is fed back to v_- through R_3 , and v_+ is connected to ground through R_4 .

$$\begin{aligned} v_+ &= v_2 \frac{R_4}{R_2 + R_4}, \\ \frac{v_1 - v_-}{R_1} &= \frac{v_- - v_o}{R_3}, \\ \frac{v_1}{R_2} - v_2 \frac{R_4}{R_2 + R_4} \frac{R_2 + R_4}{R_2 R_4} + \frac{v_o}{R_4} &= 0, \\ v_o &= \frac{R_4}{R_2} (v_2 - v_1). \end{aligned}$$

When $R_1 = R_2 = R_3 = R_4$:

$$v_o = (v_2 - v_1).$$

xi Comparator

A comparator is an op amp with v_+ and v_- connected to the voltage sources to be compared, in which v_o is high when $v_+ > v_-$ and low when $v_+ < v_-$.

xii Inverting integrator

An inverting integrator consists of an op amp where input voltage v_i is connected to v_- through an input resistor R_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback capacitor C_f .

Time domain:

$$\begin{aligned} i_i = \frac{v_i}{R_i} &= i_f = -C_f \frac{dv_o}{dt}, \\ v_o &= v_o(0^-) - \int_0^t \frac{v_i(\tau)}{R_i C_f} d\tau. \end{aligned}$$

Laplace domain:

$$\begin{aligned} I_i = \frac{V_i}{R_i} &= I_f = -C_f s V_o + C_f v_o(0^-), \\ V_o &= \frac{v_o(0^-)}{s} - \frac{V_i}{R_i C_f s}. \end{aligned}$$

xiii Inverting differentiator

An inverting differentiator consists of an op amp where input voltage v_i is connected to v_- through an input capacitor C_i , v_+ is connected to ground, and v_o is fed back to v_- through a feedback resistor R_f .

Time domain:

$$i_i = C_i \frac{dv_i}{dt} = i_f = \frac{-v_o}{R_f},$$
$$v_o = R_f C_i \left(v_i(0^-) - \int_0^t v_i(\tau) d\tau \right).$$

Laplace domain:

$$I_i = C_i s V_i - C_i v_i(0^-) = I_f = \frac{-V_o}{R_f},$$
$$V_o = R_f C_i (v_i(0^-) - s V_i).$$

xiv Active filter

Given a (passive or active) filter v_f of an input v_s and an inverting amplifier, noninverting amplifier, or unit gain buffer enforcing $v_o = K v_i$, by connecting the ground and v_i of the amplifier to the two terminals of the v_f , the two terminals ground and v_o become an active filter of v_s enforcing $v_o = K v_f$.